Lecture 2
Linux

Comp 206 – Introduction to Software Systems – Fall 2026

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September 3, 2026

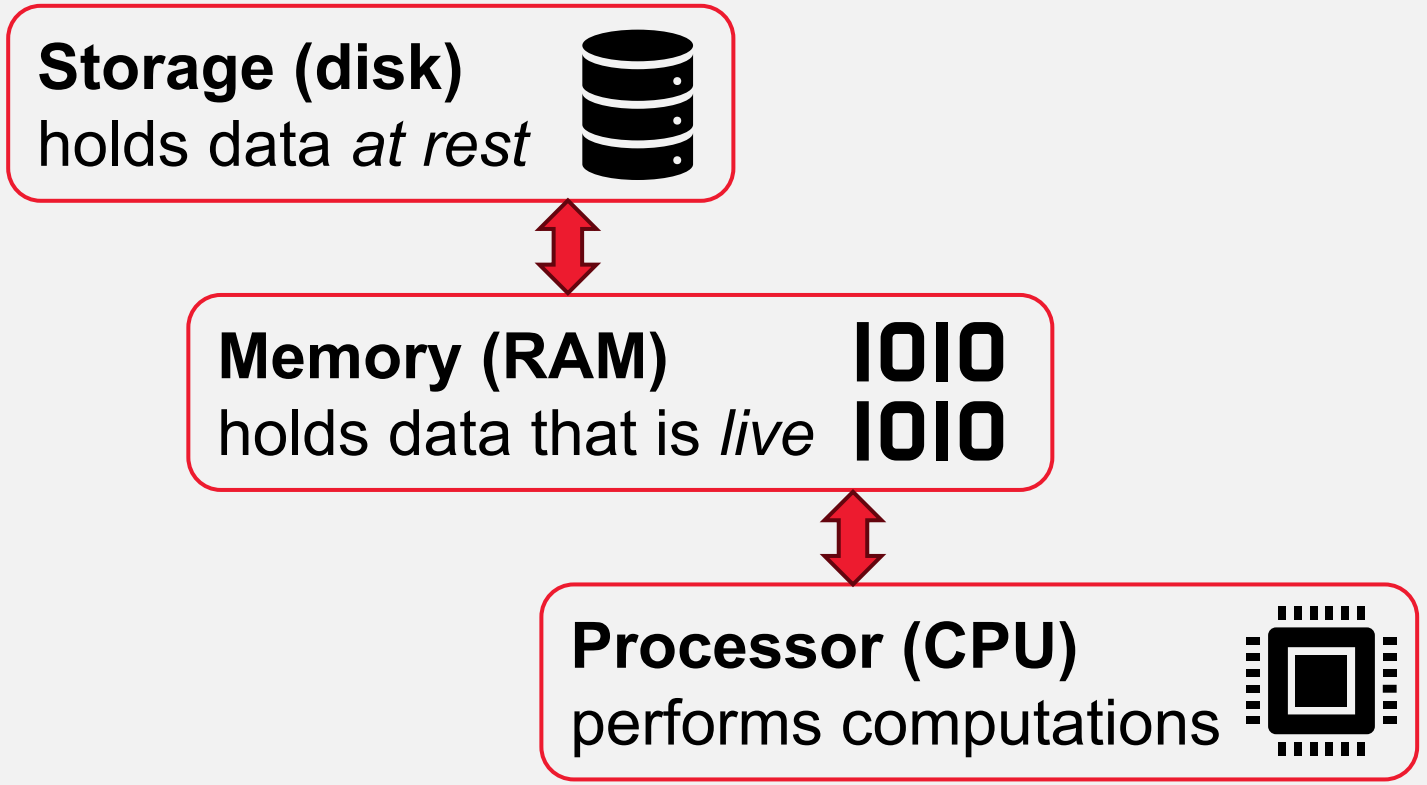
Learning goals

After this lecture, you will:

- Be familiar with the basic computer hardware components
- Be familiar with Operating Systems and their purpose
- Be familiar with the history of Unix and Linux
- Know the structure of the Linux File System
  - Including the difference between absolute vs relative values
- Have seen basic file manipulation on the Linux command line

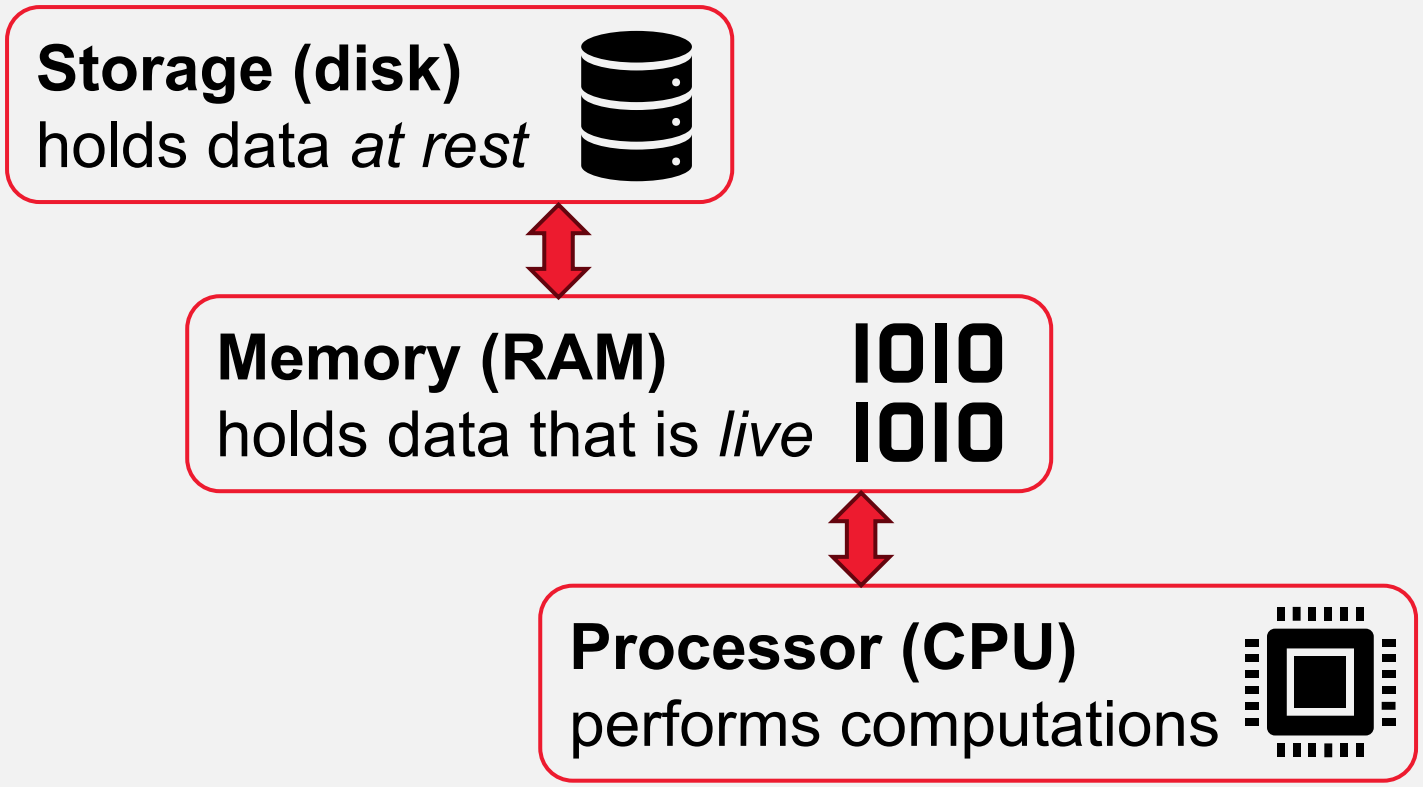
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Main computer hardware components



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Main computer hardware components




```
a = x + y
```

1. Code *Instruction* copied from RAM into a CPU register
2. Decodes as *add* instruction
3. Copies values of x and y from RAM into other CPU registers
4. Performs addition, storing result in another CPU register
5. Copies value from that register to RAM

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Interactions



User



Applications

Storage (disk)
holds data *at rest*

Memory (RAM)
holds data that is *live*

1010
1010

Processor (CPU)
performs computations







Output devices

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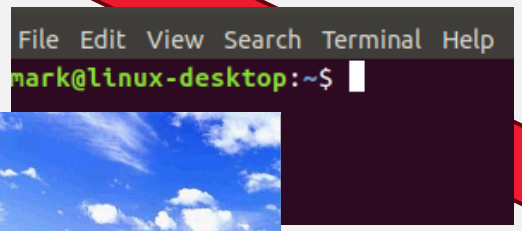
Interactions



User

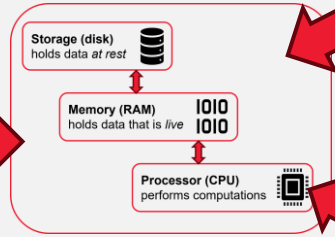


Input devices



Graphical user interface (GUI) or command-line

Common interface



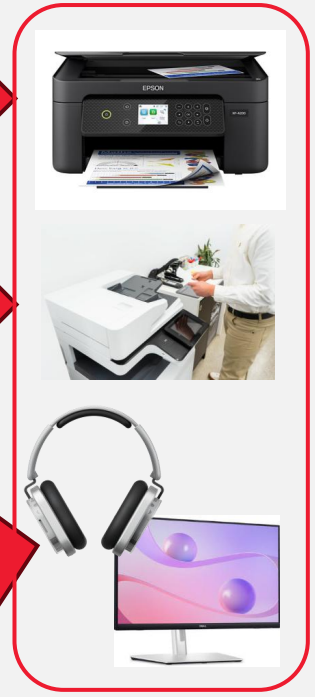
Memory & Processor

Printer manager

Custom driver

Custom driver

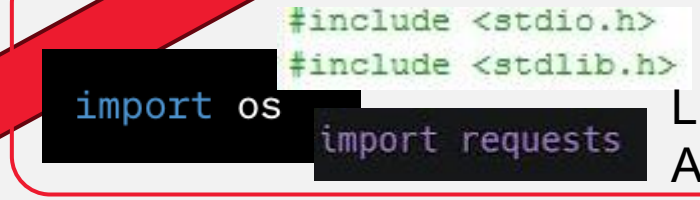
Other managers and drivers



Output devices



Applications



Libraries, APIs…

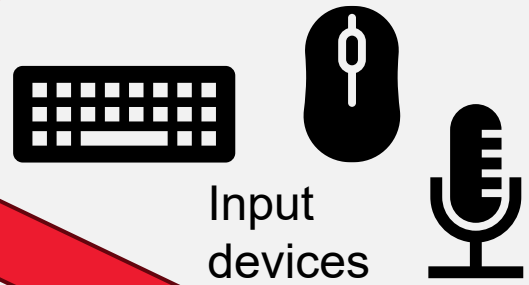
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Operating system

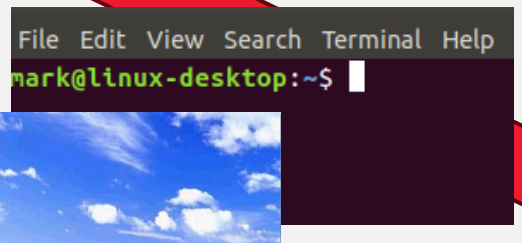
Manages and abstracts these interactions



User

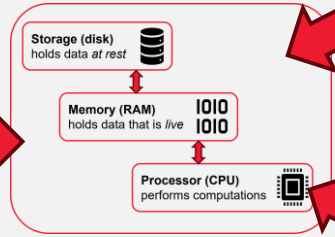


Input devices



Graphical user interface (GUI) or command-line

Common interface



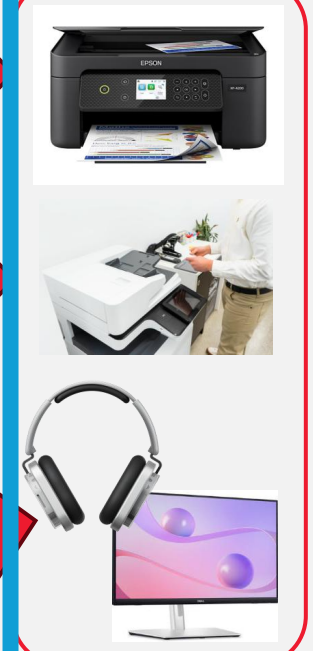
Memory & Processor

Printer manager

Custom driver

Custom driver

Other managers and drivers



Output devices



Applications



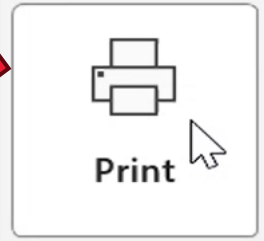
Libraries, APIs…

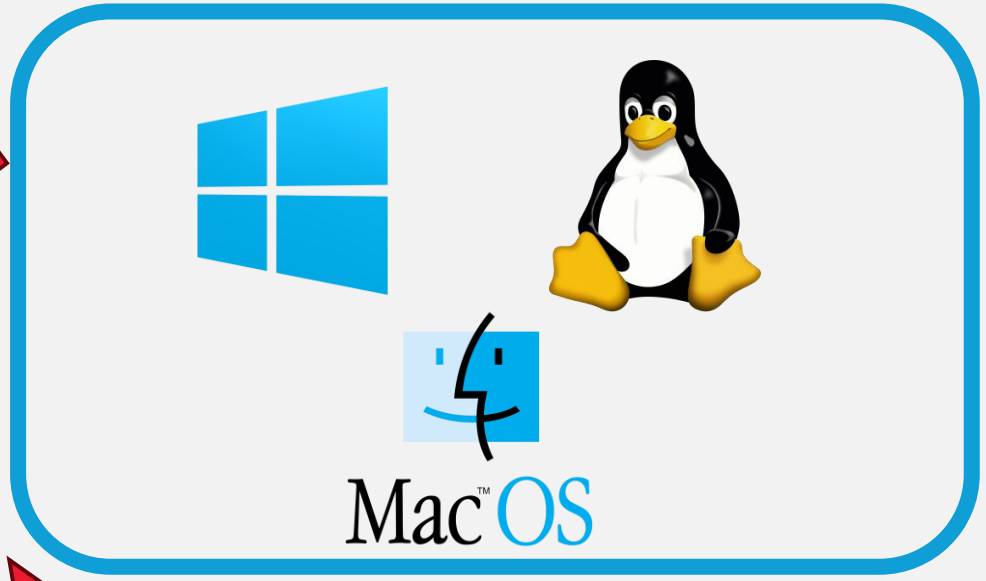
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Operating systems



User







Printer



Program

```
import os

os.startfile("C:/Users/TestFile.txt", "print")
```

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Operating system

- "Insulates" applications running on a computer, both from the *hardware* of that computer and also *from each other.*
- "Insulates" the *user* from the complex inner workings of the computer
- OS is itself a system! Components include: the *kernel*, the network stack, the filesystem, a hardware abstraction layer
- OS provides programmers with *system libraries* to perform OS-level operations, e.g. using network connections, allocating memory, managing files, using peripherals (Bluetooth, USB, etc.)
- **Primary function of the OS**: managing the computer's resources
  - Multiple applications and users want to use processing time and space at the same time! OS handles sharing of these resources.

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Linux

- A family of free and open-source operating systems (called Linux distributions) based on the Linux kernel
  - Kernel: performs core OS functions. Includes: networking, filesystem, user accounts/permissions, memory management, device drivers, and even more
  - OS components that are not part of the kernel include the user interface (shell, GUI), the utilities, the bootloader, etc.
- Most distributions are based on a combination of the Linux kernel and utilities from the GNU project:
  - Ubuntu, Debian, CentOS, Red Hat Enterprise Linux, Arch Linux, Kali Linux, and many, many more
  - Other distributions not based on GNU include Android and ChromeOS

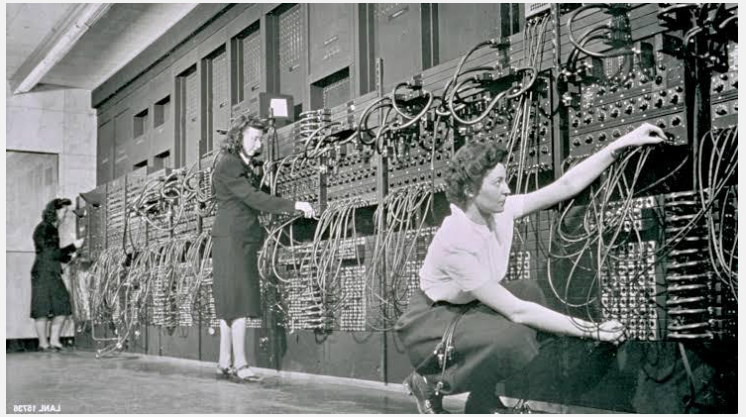
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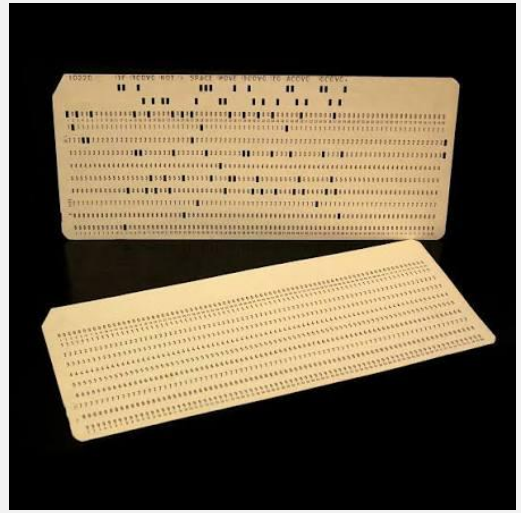
Quick history of Linux

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Early computing

- “Computer” is a job title. People, mostly women, do calculations by hand.
- The first electronic computers are built in this decade. People reprogrammed these room-sized machines manually by rewiring them. This could take weeks.
  - Role of the human computers shifts from doing calculations to operating these machines
- Punched cards start being used to store data and instructions





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Road to modern OSes

- Libraries
  - Common punched cards shared
- Batch Processing
  - Don’t have to input punched cards one by one
- Single OS for multiple machines
  - Don’t need a different OS for each different computer!
- Multiprogramming
  - Run multiple programs at same time by switching between them
- Time sharing
  - Allow multiple users to use computer
  - First attempt at general-purpose time-sharing system: CTSS (MIT, 1961)



62500 punched cards:
4.5 megabytes of data (1955)

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Towards Unix

- Successor of CTSS: **Multics** (MIT, GE & Bell Labs, 1969)
  - Many important new ideas: Hierarchical file system, I/O redirection, access-control lists, shell, etc.
  - Complex and big, poor performance: Bell Labs quits the project
- Ken Thompson, a Bell Labs scientist working on this project, wrote a game called *Space Travel* on Multics
- When the project was cancelled, he decided to port the game to the PDP-7 computer, along with Dennis Ritchie

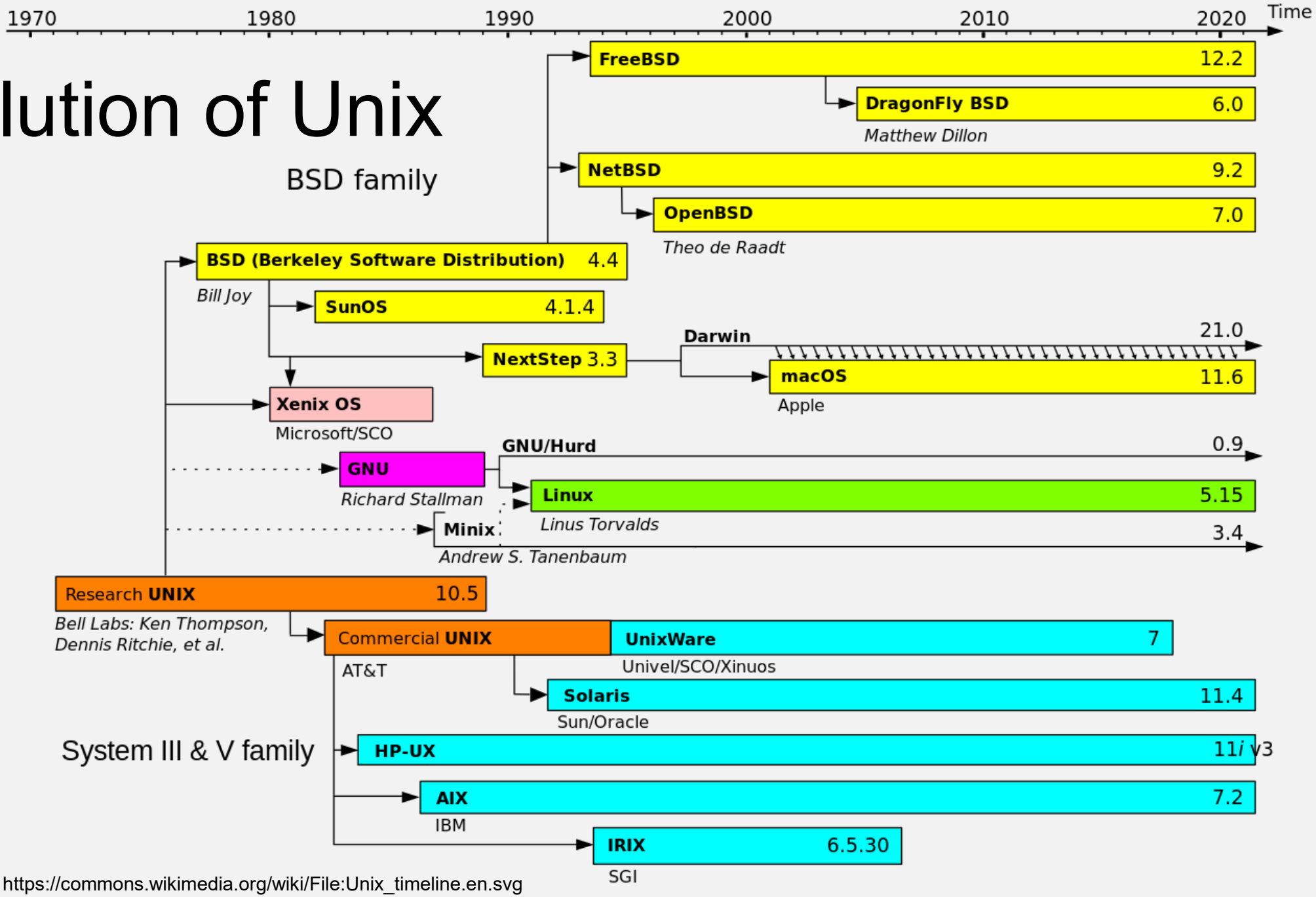
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Beginnings of Unix

- The last researchers from Bell Labs to work on Multics enjoyed many of its core ideas despite its failure
- The tediousness of writing *Space Travel* on the PDP-7 convinced Thompson and Ritchie to start working on a new operating system for it: **Unix** (1969)
  - Brian Kernighan, Douglas McIlroy, and Joe Ossanna are the other developers
- Unix presented by some of its developers in 1982: youtu.be/XvDZLjaCJuw

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Evolution of Unix



https://commons.wikimedia.org/wiki/File:Unix_timeline.en.svg

1991: Linux

```
From: torvalds@klaava.Helsinki.FI (Linus Benedict Torvalds)
Newsgroups: comp.os.minix
Subject: What would you like to see most in minix?
Summary: small poll for my new operating system
Message-ID:
Date: 25 Aug 91 20:57:08 GMT
Organization: University of Helsinki


Hello everybody out there using minix -

I'm doing a (free) operating system (just a hobby, won't be big and
professional like gnu) for 386(486) AT clones.  This has been brewing
since april, and is starting to get ready.  I'd like any feedback on
things people like/dislike in minix, as my OS resembles it somewhat
(same physical layout of the file-system (due to practical reasons)
among other things).

I've currently ported bash(1.08) and gcc(1.40), and things seem to work.
This implies that I'll get something practical within a few months, and
I'd like to know what features most people would want.  Any suggestions
are welcome, but I won't promise I'll implement them :-)

              Linus (torvalds@kruuna.helsinki.fi)

PS.  Yes - it's free of any minix code, and it has a multi-threaded fs.
It is NOT protable (uses 386 task switching etc), and it probably never
will support anything other than AT-harddisks, as that's all I have :-(.

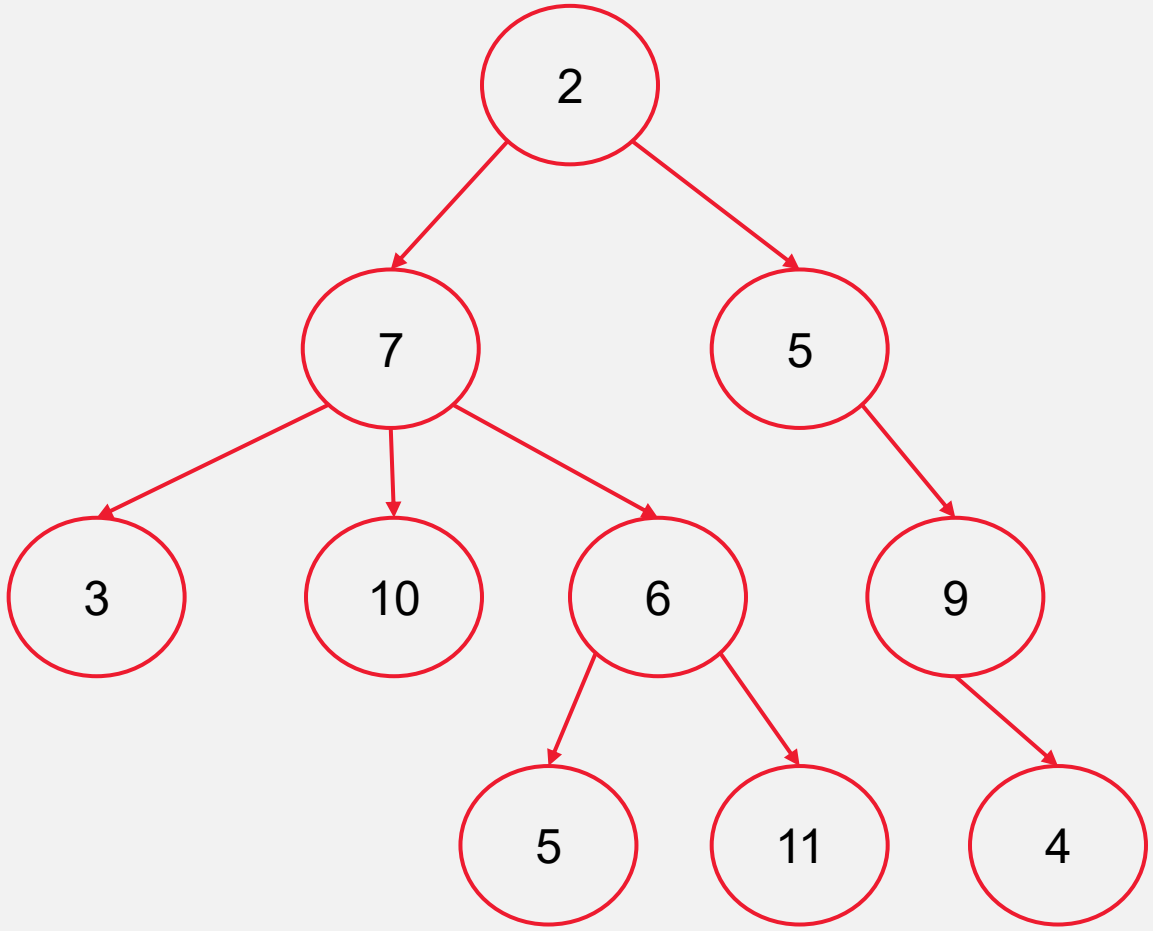
 Judging from the post, 0.01 wasn't actually out yet, but it's close.
 I'd guess the first version went out in the middle of September -91. I
 got some responses to this (most by mail, which I haven't saved), and
 I even got a few mails asking to be beta-testers for linux. After that
 just a few general answers to quesions on the net:
```

File System (FS)

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Underlying concept: Tree data structure

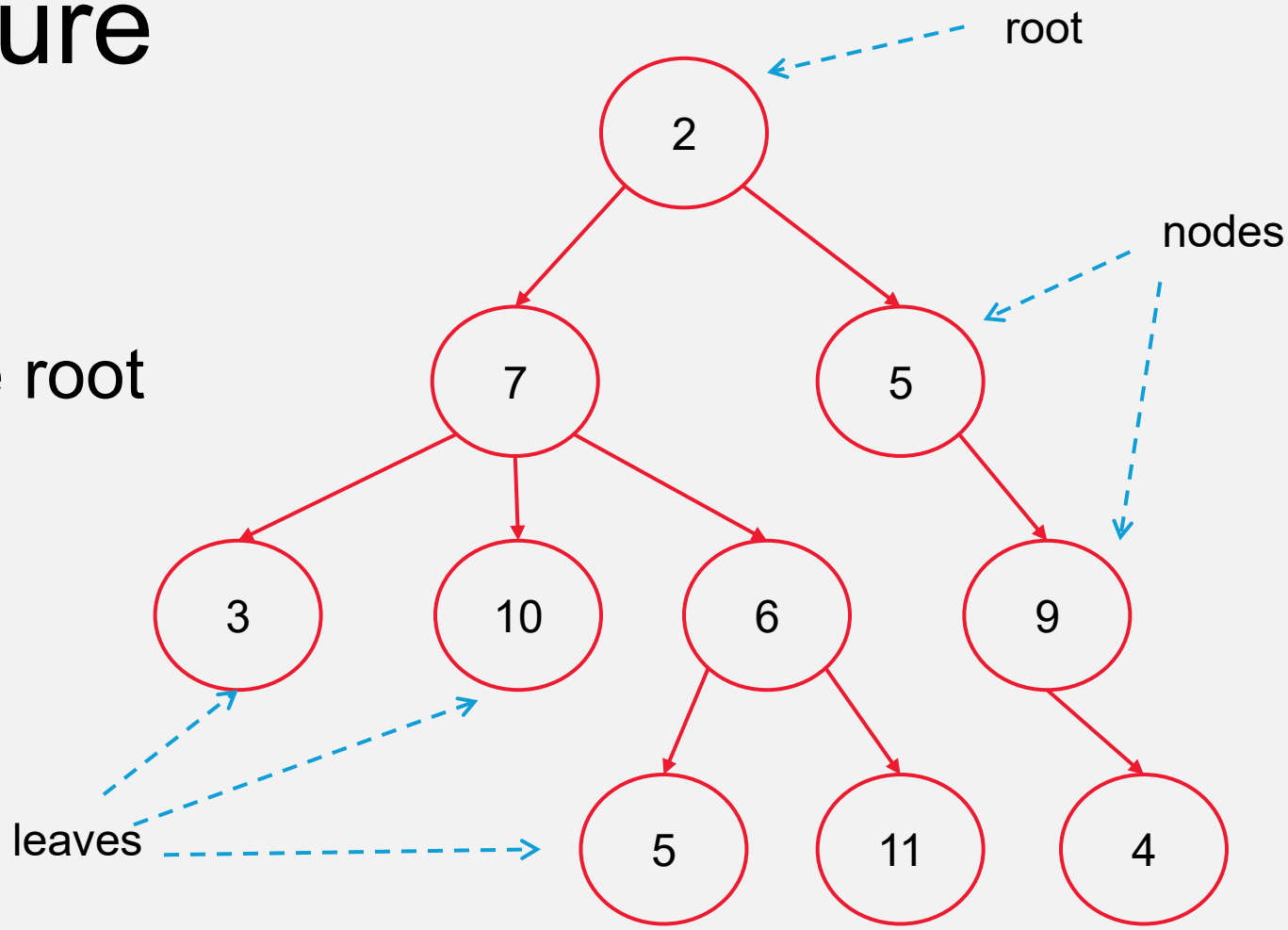
- Node vs leaf
- Data stored in nodes
- Distinguished node: the root



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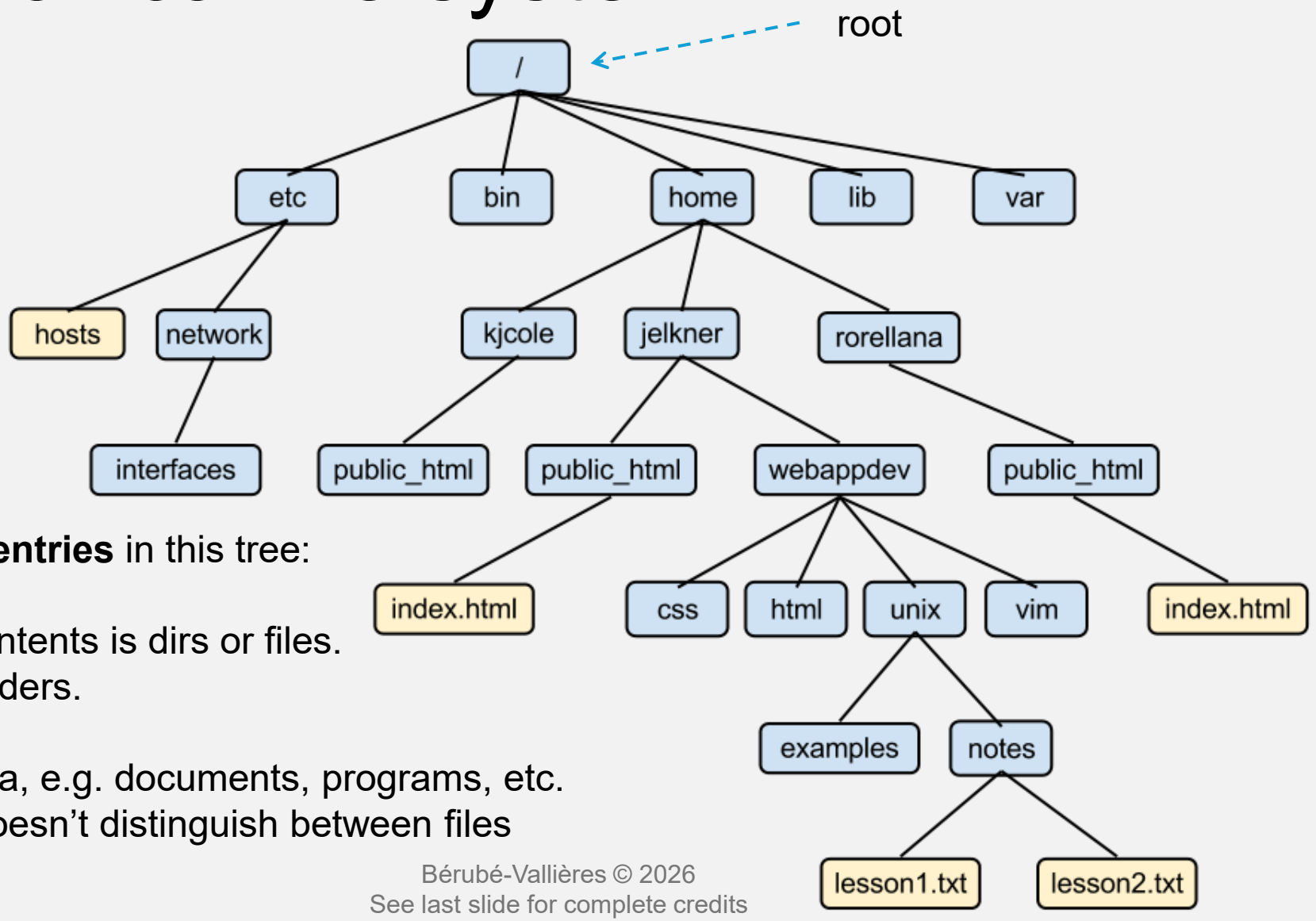
Underlying concept: Tree data structure

- Node vs leaf
- Data stored in nodes
- Distinguished node: the root



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Hierarchical file system:

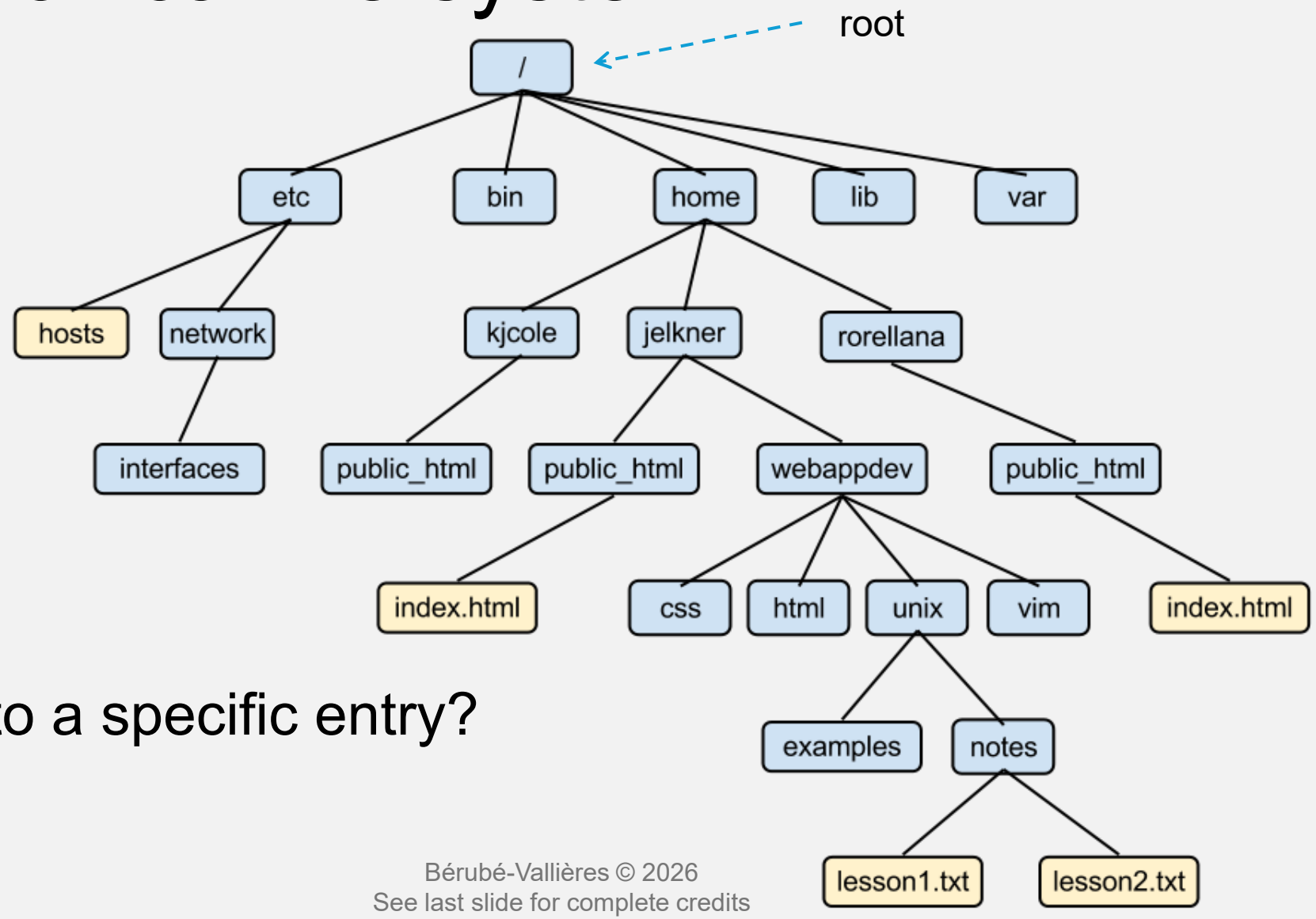


Two different kind of **entries** in this tree:

- **Directories**
  - Structural. Contents is dirs or files. Also called folders.
- **Files**
  - Content is data, e.g. documents, programs, etc. File system doesn't distinguish between files

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Hierarchical file system:



How to refer to a specific entry?

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Intuition: How to refer to a location in space?

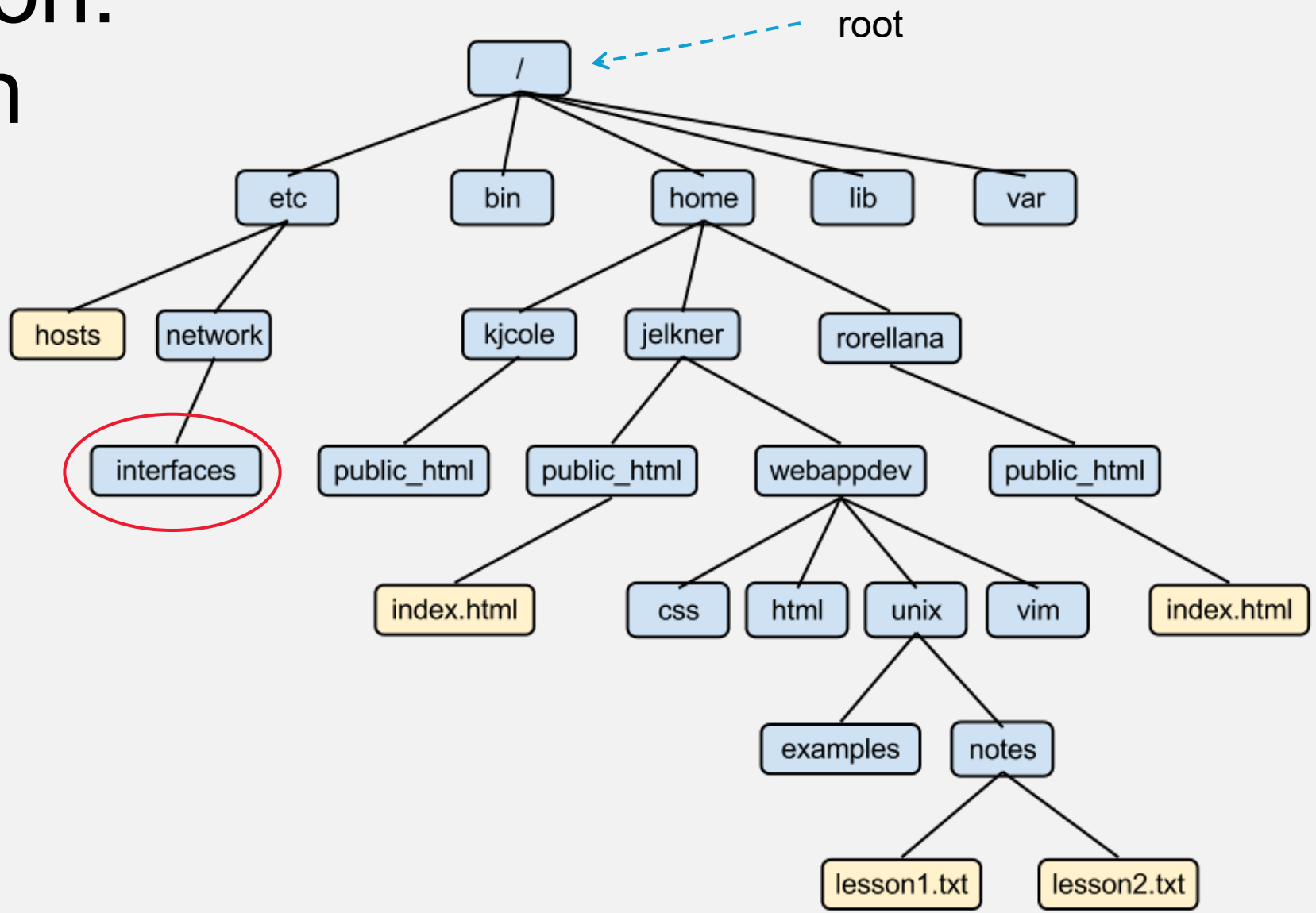
- **Absolute** approach, the full address:
  - Room 328, McConnell Eng. Bldg., 3480 rue University, Montreal, Quebec, Canada
  - Doesn't depend on current location!

- **Relative** approach, directions from "here":
  - Down the hall, on the left
  - Depends on current location: this doesn't work if you're saying this from another building!

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File/dir location: absolute path

Example: location of 'interfaces' directory



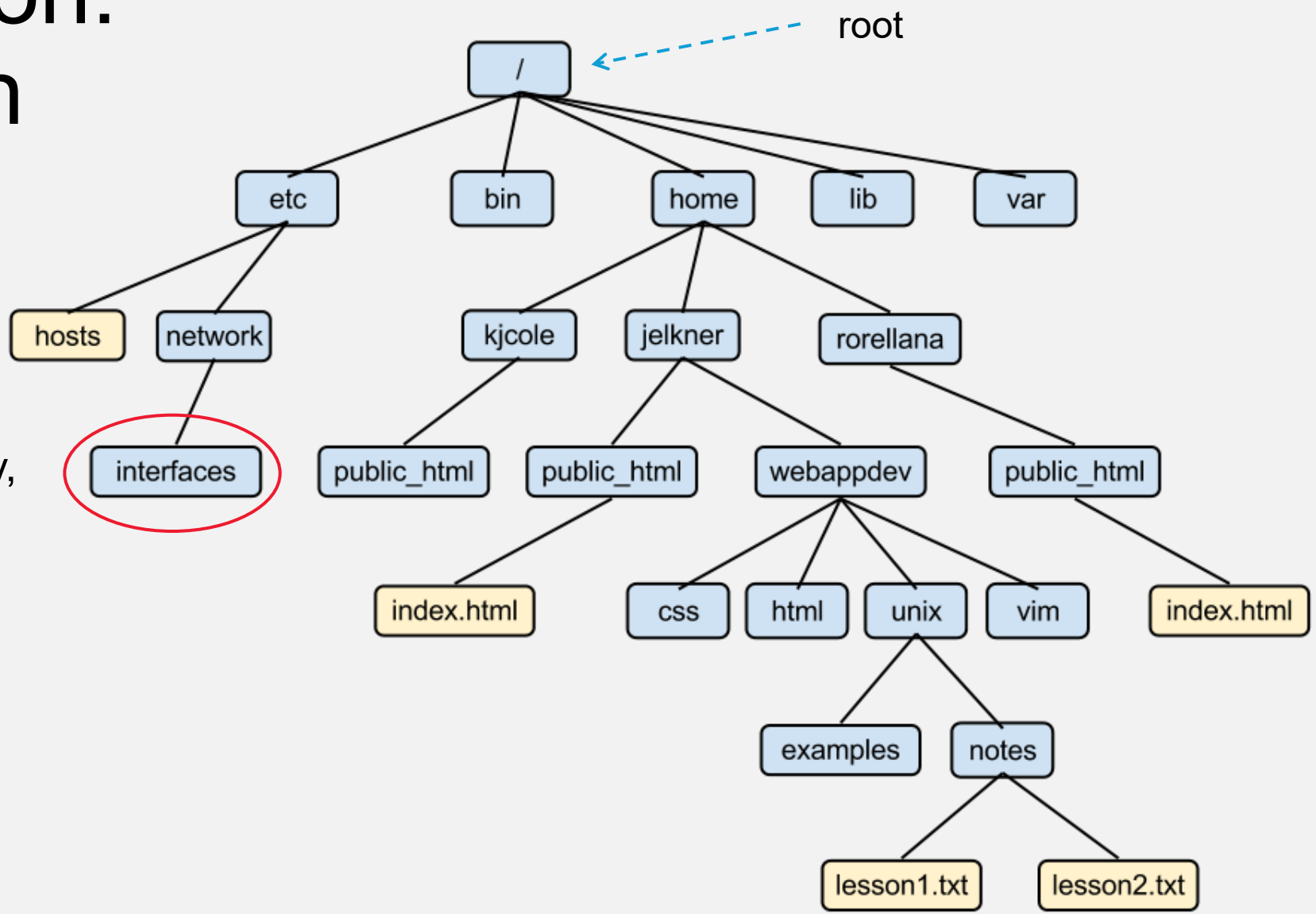
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File/dir location: absolute path

Example: location of 'interfaces' directory

Absolute approach:

- Start from the root directory,
- Go through 'etc'
- Go through 'network'
- Here is 'interfaces'



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File/dir location: absolute path

Example: location of 'interfaces' directory

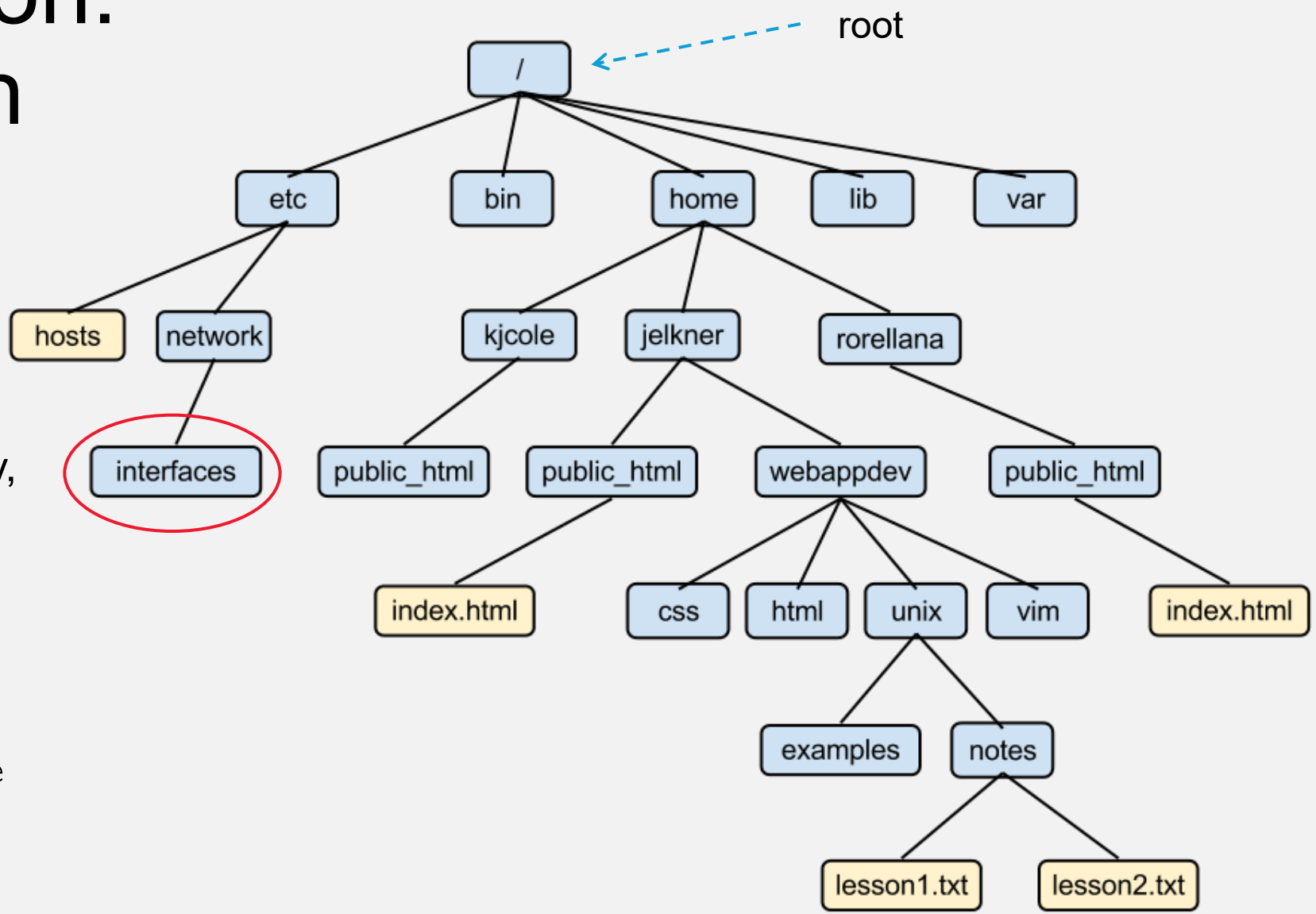
Absolute approach:

- Start from the root directory,
- Go through 'etc'
- Go through 'network'
- Here is 'interfaces'

Absolute path:

`/etc/network/interface`

Begins with a slash (/) !

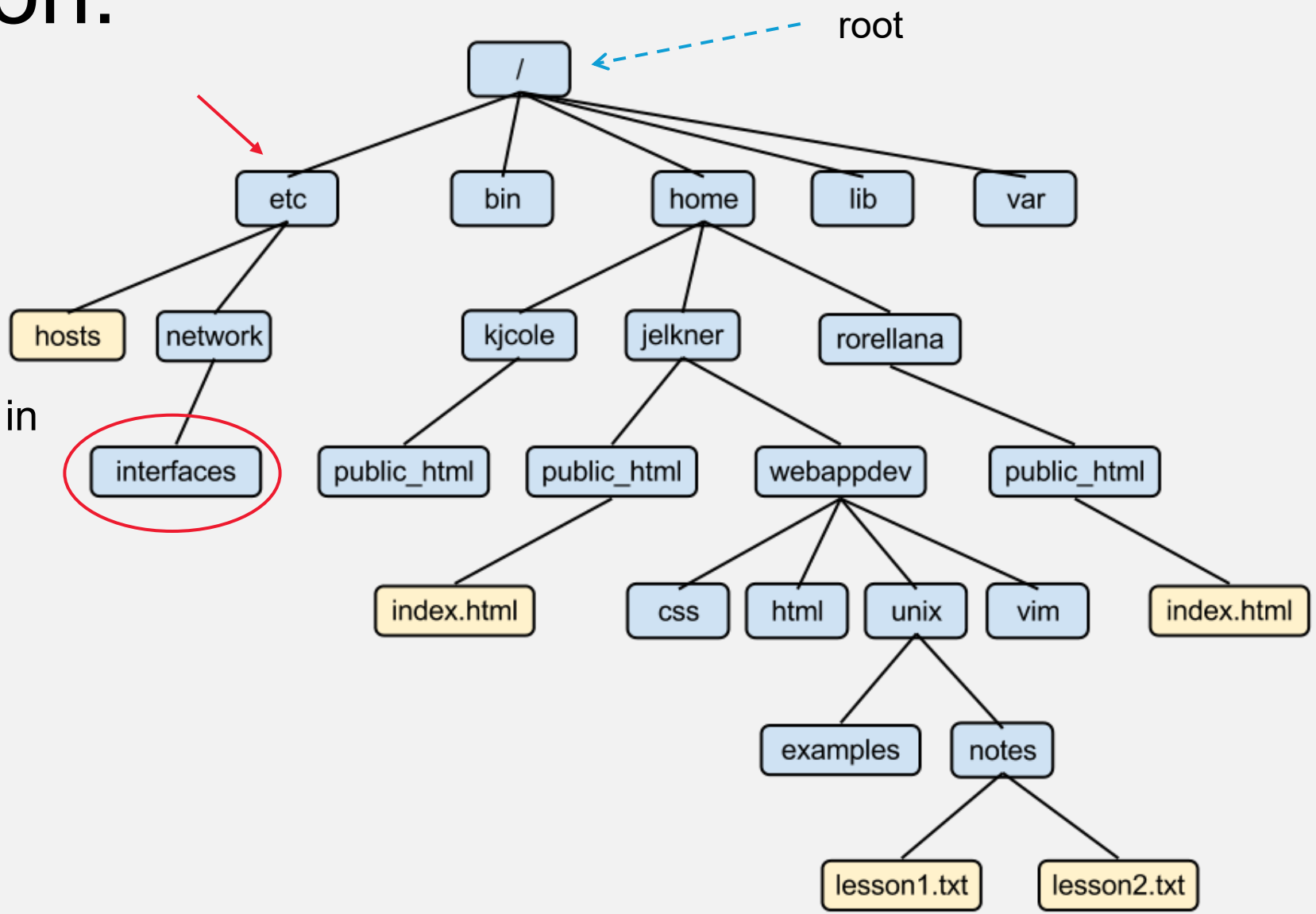


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File/dir location: relative path

Example: location of 'interfaces' directory

Assume that we are currently in directory 'etc'



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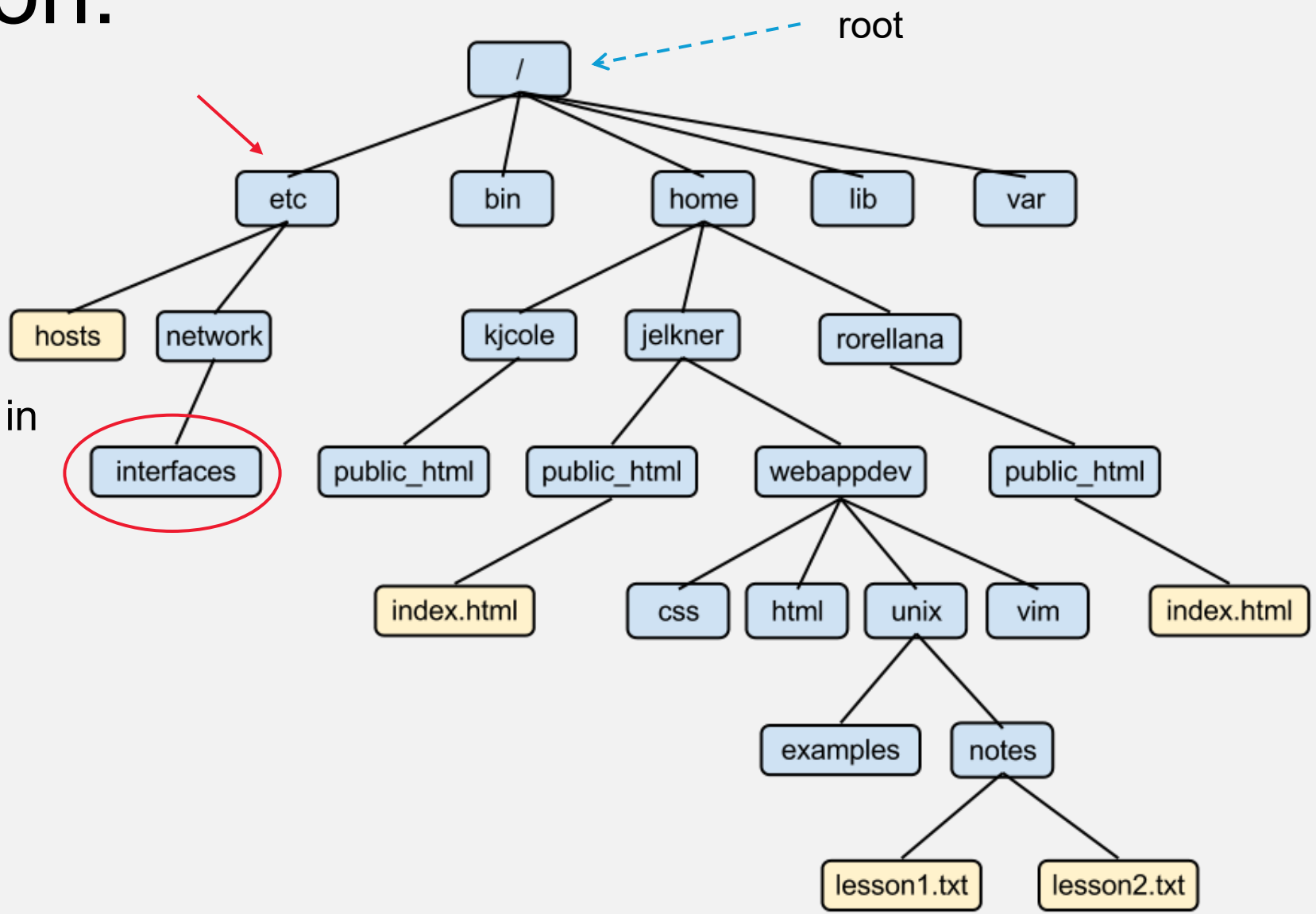
File/dir location: relative path

Example: location of 'interfaces' directory

Assume that we are currently in directory 'etc'

Relative approach:

- From here (assume 'etc')
- Go through 'network'
- Here is 'interfaces'



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File/dir location: relative path

Example: location of 'interfaces' directory

Assume that we are currently in directory 'etc'

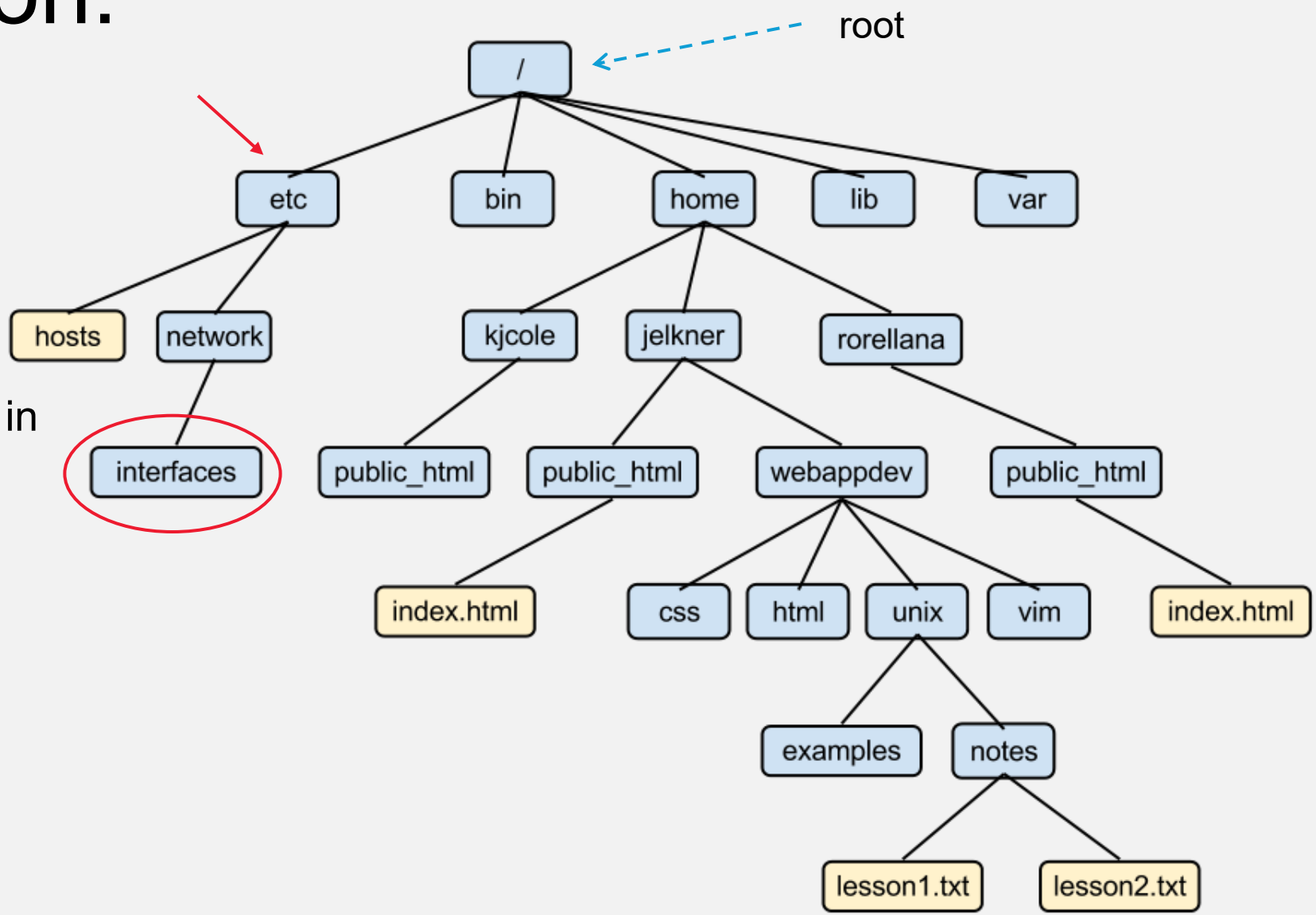
Relative approach:

- From here (assume 'etc')
- Go through 'network'
- Here is 'interfaces'

Relative path:

`network/interface`

Does not begin with a slash!

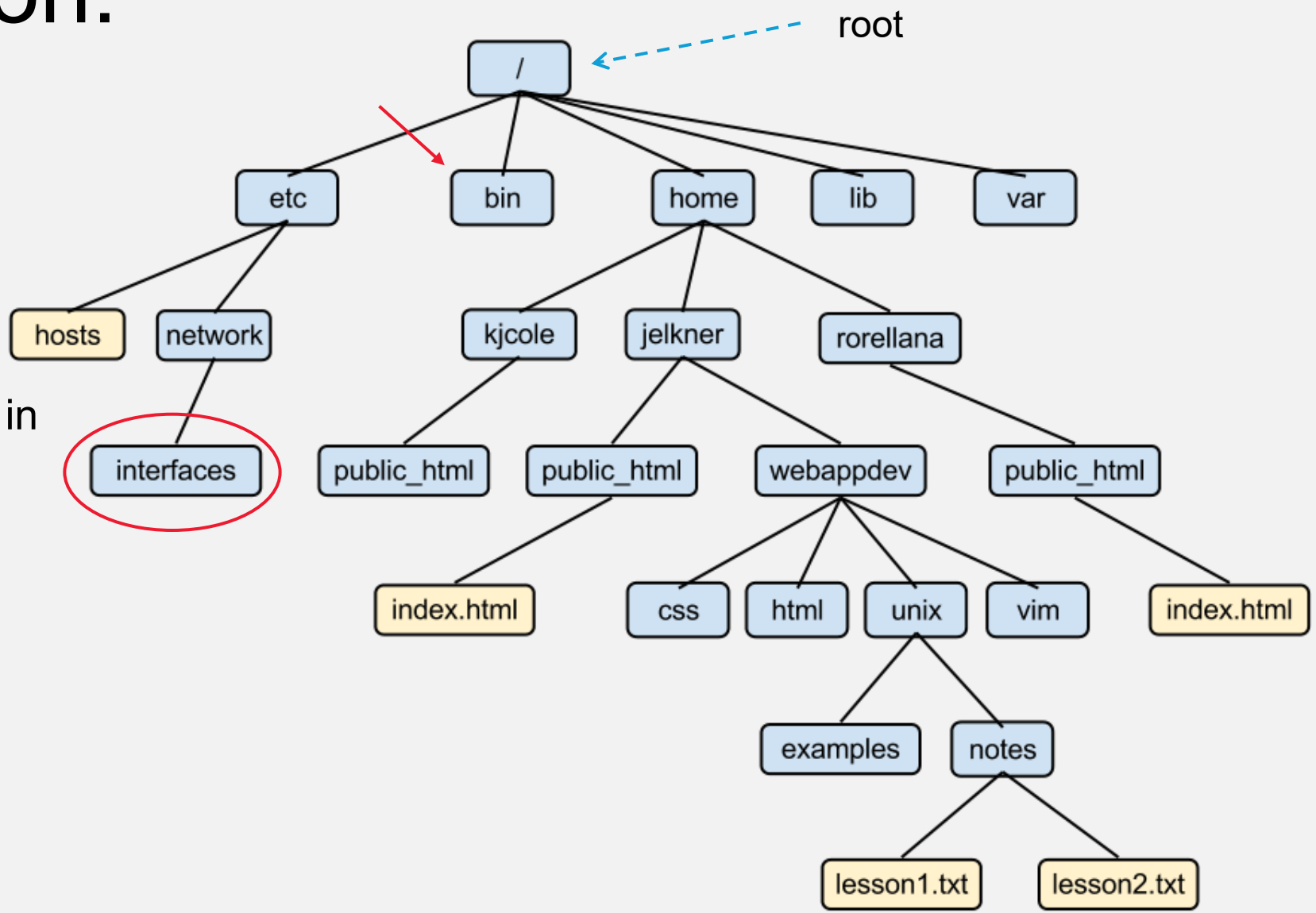


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File/dir location: relative path

Example: location of 'interfaces' directory

Assume that we are currently in directory 'bin'



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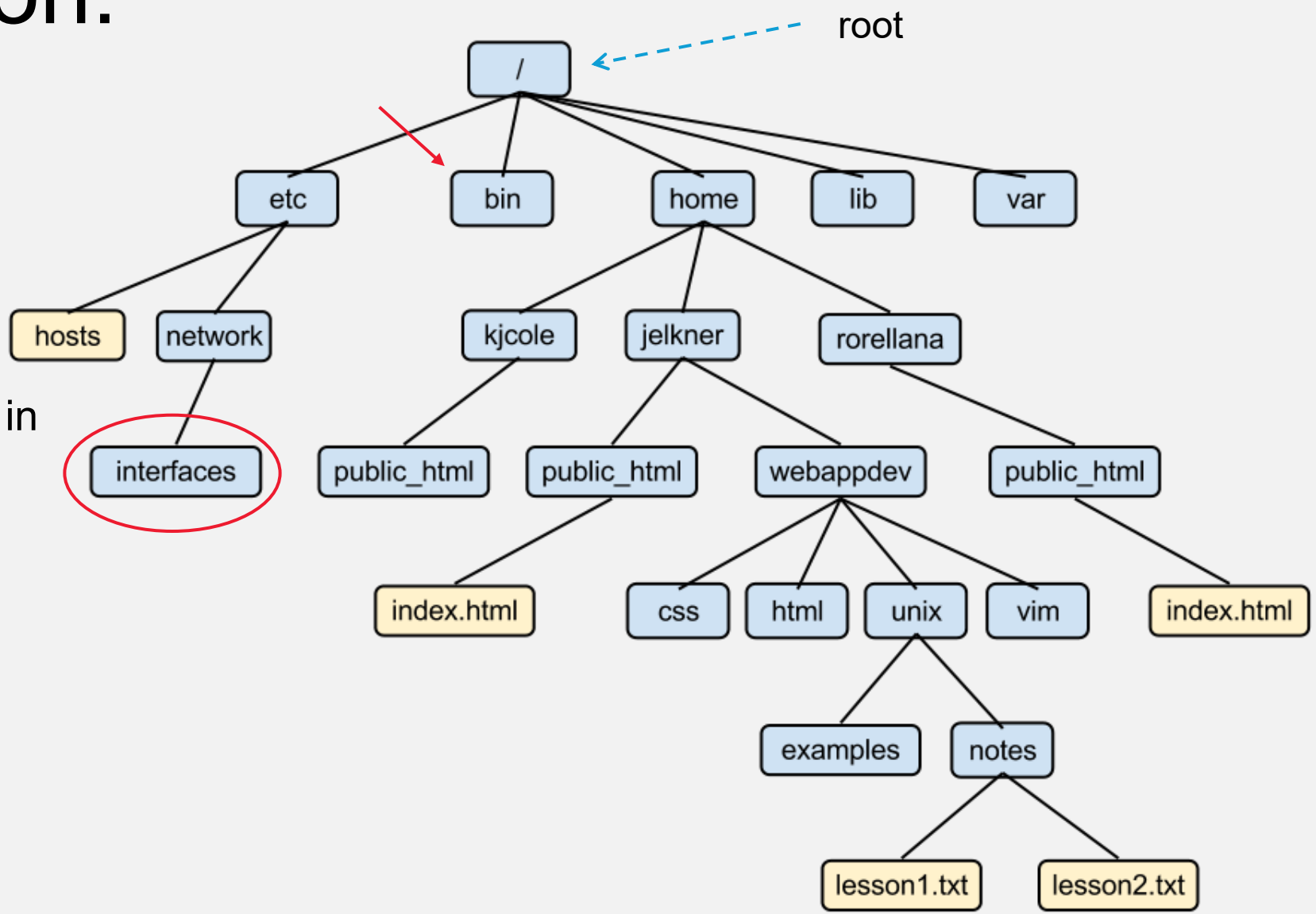
File/dir location: relative path

Example: location of 'interfaces' directory

Assume that we are currently in directory 'bin'

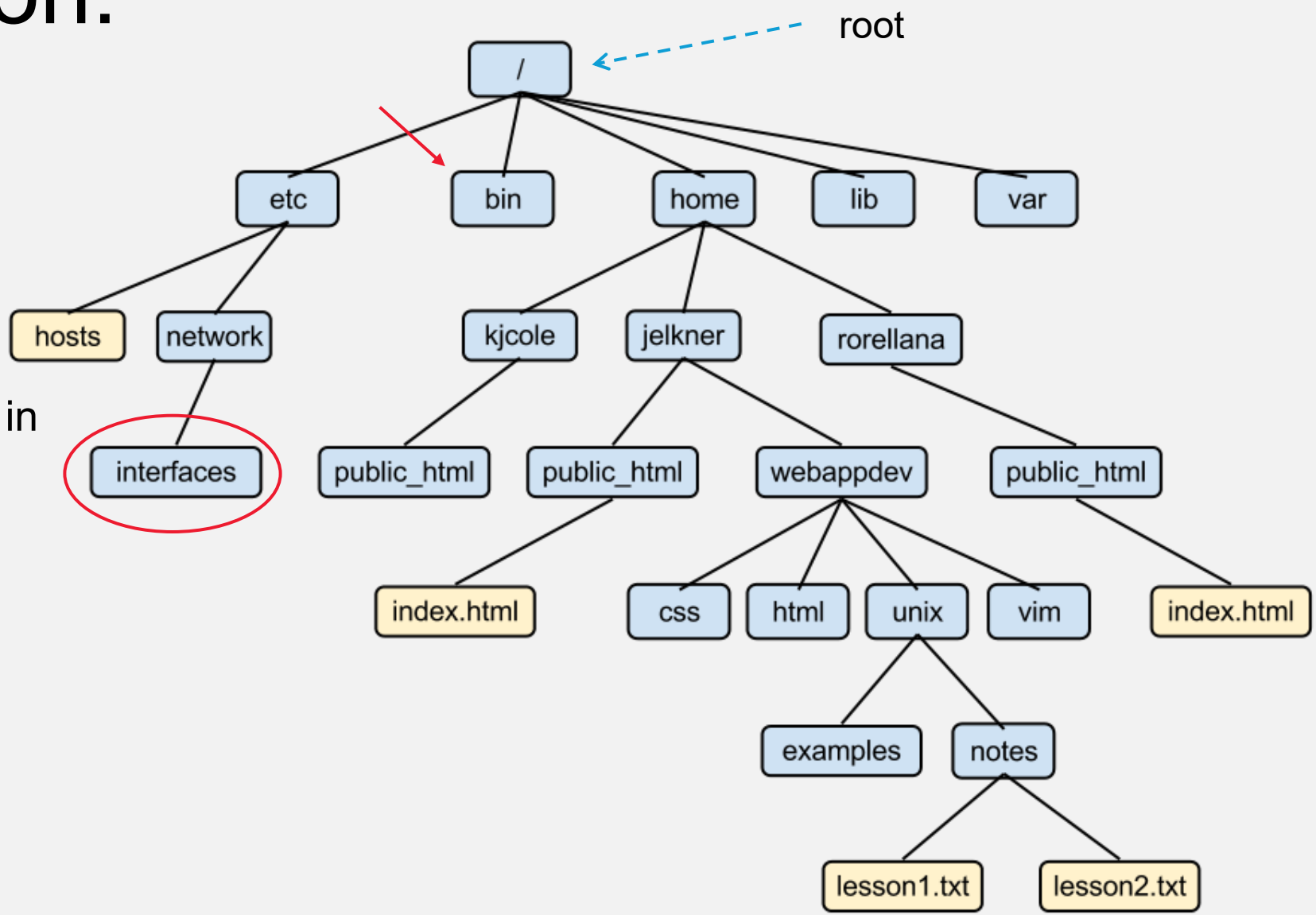
Relative approach:

- From here (assume 'bin')
- Go up
- Go through 'etc'
- Go through 'network'
- Here is 'interfaces'



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File/dir location: relative path



Example: location of 'interfaces' directory

Assume that we are currently in directory 'bin'

Relative approach:

- From here (assume 'bin')
- Go up
- Go through 'etc'
- Go through 'network'
- Here is 'interfaces'

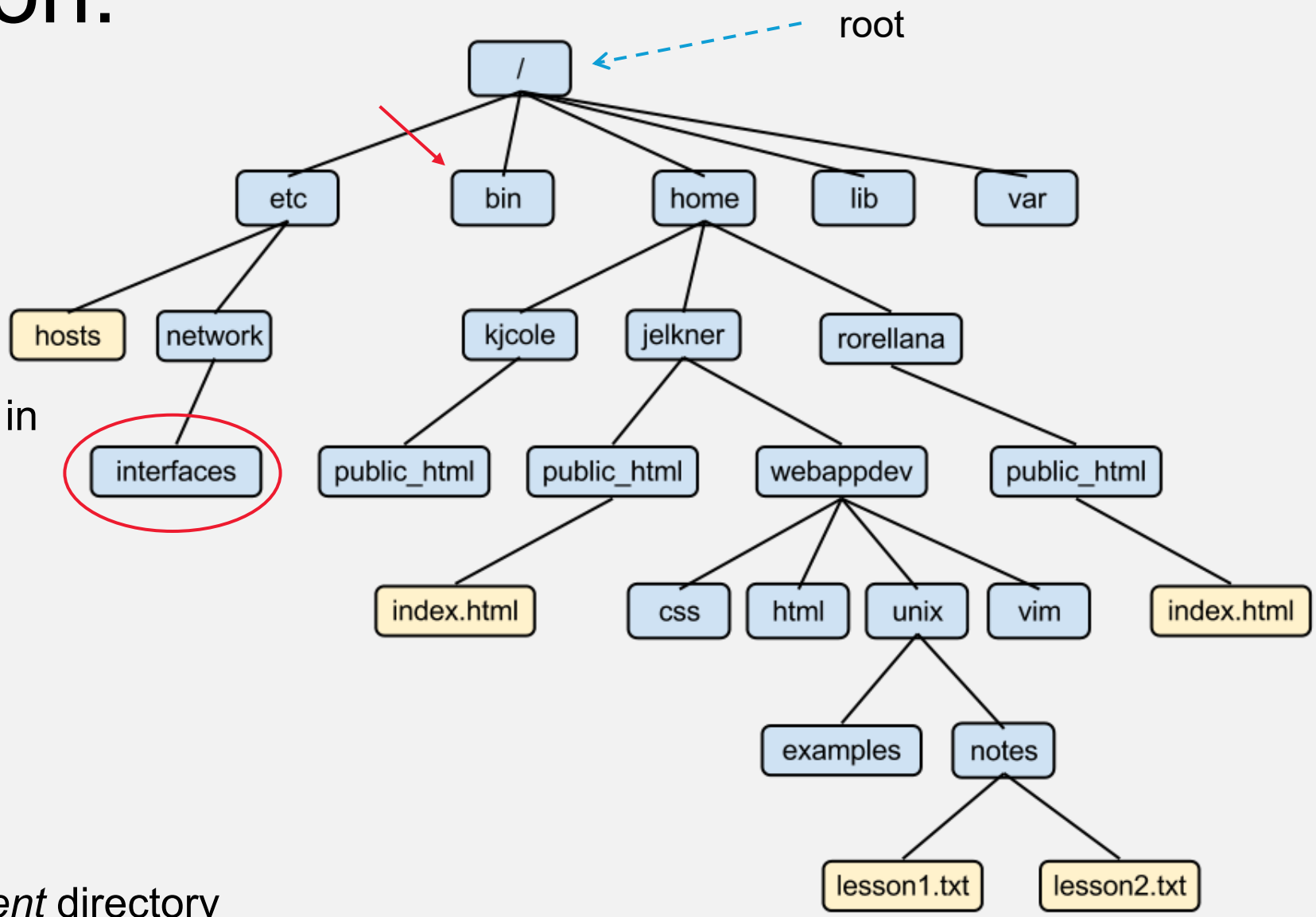
Relative path:

`../etc/network/interfaces`

Does not begin with a slash!

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File/dir location: relative path



Example: location of 'interfaces' directory

Assume that we are currently in directory 'bin'

Relative approach:

- From here (assume 'bin')
- Go up
- Go through 'etc'
- Go through 'network'
- Here is 'interfaces'

Parent directory

Relative path:

`../etc/network/interfaces`

Does not begin with a slash!

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Demo

Here is an early look at commands we will learn next week!

- Using touch to create a file
- Using mkdir to make a directory
- Using rm to remove a file
  - And rm -r to remove a directory
- Using mv to move a file/directory
- Using cp to copy a file
  - And cp –r to copy a directory

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Additional credits

- Some content from this lecture is based on:
  - Past Comp 206 slides from Prof. Vybihal, Kopinsky and Errington

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